

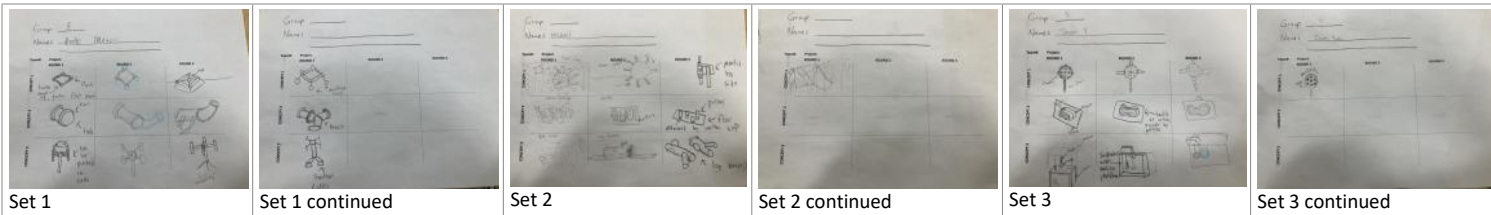
Element D student work

Thursday, September 25, 2025 12:55 PM

Idea Generation and Iteration

First Round and Iterations

To start the idea generation process, our group, in conjunction with a solo member of another group, drew and iterated upon a multitude of ideas for potential devices with which the turtle could interact. The product of this activity was 9 general concepts, with a total of 31 iterations.



Second Round: Narrowing Down Ideas

The next step from this point is to narrow down the list of potential ideas in order to approach the final pick. To do this, we will compare each idea to the 5 criteria listed in Element C, one by one, and eliminate any concepts which do not comply with the criteria.

In the name of being concise, the concepts will be named with three numbers, representing set, concept, and round respectively (1.2.3, for instance, is the third iteration of the second concept in the first set of concepts)

1. Must not harm the turtles: This criteria eliminates 1.1.2-1.1.4, 1.2.2-1.2.4, 2.1.4, 2.2.1 and 2.2.2, 2.3.3, and all of 3.3. These concepts were all eliminated because they have the potential to trap the turtle underwater, which will result in harm and eventually death.
2. Species/environment appropriate enrichment: This criteria eliminates all of 2.1, as these designs enrich the turtle by rewarding them with food. According to field experts, the turtles are currently overweight, so a form of enrichment which relies on food is inappropriate.
3. Durability and cleanability: The durability criteria eliminates the 1.3 series of designs, as such a device is going to be somewhat fragile, with components that could be damaged by the turtles. Cleanability doesn't eliminate any surviving competitors.
4. Environmental compatibility: This criteria eliminates 2.1.1 (a tree branch), which is too large to place inside of a 4x1x2 foot tank. Also, 2.2.3 is eliminated, because it requires the bottom of the tank to be smooth to mount with suction cups, which is not the case in this tank.
5. Variety and stimulation: This criteria does not eliminate any concepts, as all surviving ideas are capable of providing various forms of mental stimulation to the turtles.

Third Round: Elaborating on Contenting Ideas

Four ideas have survived initial contact with the criteria listed in Element C. The following list names each idea properly, and elaborates on their designs a bit.

- 1.1.1: the Pool Float: The pool float consists of a square made out of some kind of hollow pipe, likely PVC, joined at all four corners using pipe corner pieces, which floats on top of the water. The straight sections of pipe have rope coiled around them to soften the surfaces and add texture. The idea behind this design is that the turtles can swim up inside of or next to the float, and push it around or rub their shells on the textured rope. The design will be approximately as wide as the widest turtle, so that both turtles can use the device.
- 1.2.1: the Tube: The tube is the simplest of the designs, consisting of a singular piece of short, wide pipe which the turtles can swim and rest in. The pipe is held down by partially burying it in the existing pebbles on the floor of the tank. The pipe must be wide enough so that the largest turtle can easily fit inside, and has no chance of getting stuck.
- 3.1.1: the Sea Mine: The sea mine involves a buoyant, spiked ball, on some sort of string or chain to hold it down. The idea is for the turtles to bump it or bat at it, while being able to bite it without damaging it. The preferred material for the ball is likely to be some sort of waterproof rubber, which is flexible, soft, and durable enough to be bitten without breaking or harming the turtle. The chain is connected to a wide base which is buried under bedding pebbles. Size can vary, but will likely be about 3 inches in diameter, which is too large to be eaten, but too small to take up a large amount of space.
- 3.2.3: the Movable Brush: the movable brush is a brush which is attached to the outside of the cage via a magnet and friction. A magnetic handle is placed on the outside, pulling the brush towards it and holding it in place. However, the handle can be moved, which allows the user to interact with the turtle. When not in use by a human, the turtle may still rub against the brush, in order to clean its shell. The brush and it's housing are to be made from plastic, with thin, rigid bristles to allow the turtle a sufficiently rough surface to clean itself with. The brush itself should be roughly 5 inches by 2 inches, which creates a long and narrow profile optimal for brushing against the side of a turtle's shell.

Fourth Round: Selection of Concept to be Developed

To decide on which idea to pursue, we created the following decision matrix to weigh our options. 3 is

the highest score in a category, 1 is the lowest, and the concept with the highest score wins.

	Safety	Durability	Cleanability	Interactivity	Cost	Simplicity	Sum
Pool Float	3	2	2	3	2	2	14
Tube	1	3	1	2	3	3	13
Sea Mine	2	1	2	2	2	1	10
Brush	3	2	3	3	1	1	13

Design Choice and Justification

Following the results of the decision matrix, the Pool Float is the concept which will proceed to the prototype phase. First, however, a brief justification of how the float will meet each criteria.

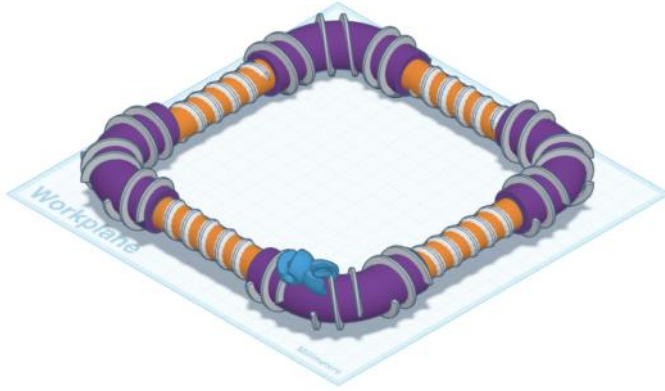
- Must not harm the turtles: The float is large enough to not trap a turtle inside of it, and is located about the surface of the water, so the turtle is not at risk of drowning. Additionally, the float is to be made out of water rated materials, which will be tested to ensure they do not degrade and/or pollute the water over an extended period of time.
- Species/environment appropriate enrichment: Both turtles are waterborne, and regularly swim around to entertain themselves. Having an object to interact with while swimming will enhance their already existing enrichment behaviors.
- Durability and cleanability: The materials used to construct the float will be durable, with no notable points of failure, such as PVC, waterproof PVC sealant, and maritime grade rope. The float is also very easy to clean, as it can be removed from the water and taken to a sink.
- Environmental compatibility: The float is sized to be as wide as the largest turtle, and given that the largest turtle fits within its tank, the float will therefore fit in as well. Additionally, the materials are chosen to reflect the aquatic nature of the environment.
- Variety and stimulation: In addition to being mobile and encouraging swimming, the wound rope provides extra texture the turtle can interact with.

Plan of Action

1. Measure largest turtle's width.
2. Use width to scale design.
3. Research types of rope, and select a type which can survive extended submersion and is bio friendly. Purchase a maritime rated rope-to-rope connector clamp to hold both sides together.
4. Research types of PVC sealants, and choose one which is waterproof and nontoxic.
5. Purchase 4 straight sections of 1 inch diameter PVC pipe, and four 90 degree corner junctions. The straight sections of PVC should be equal in length to the width of the widest turtle.
6. Fit the pipes together, applying sealant to the male side's threads before screwing them into the junctions for a waterproof fit.
7. Purchase a length of half inch diameter rope, equal to: $((SL+JL)*\pi*3)+4$ inches
 - a. Where SL is the length of a straight pipe section in inches, and JL is the length of a junction along its curve in inches.
 - b. This will give enough rope to coil around the entire float at one rotation per 2 inches.
8. Affix both sides of the rope into the clamp, ensuring it is under moderate tension, enough that the rope is held in place by friction.
9. Allow a period for the sealant to dry, in line with manufacturer directions.
10. Once the sealant is dry, place the prototype in a container of water. First, check that the prototype doesn't sink. Then, test the water for contaminants using a standard water quality test strip, then use another test for ammonia and nitrates.
11. Leave the prototype to sit for a period of 7 days. Then, test the quality of the water again. Continue repeating this test for as many weeks as possible.

Blueprint

Dimensions are not yet set in stone, as the width of the turtle is not known. However, the length of the straight pipes used will be equal to that width. To assemble, the four straightaways should be threaded into the four junction pieces, with sealant applied to the threats to make the seals watertight. Once the pipes are assembled, wrap the rope around such that each revolution takes up roughly two inches. Then, once the rope meets itself, take both ends and place them into the clamp.



Note from Mr Martz – excellent! Thorough! Well articulated!
So many good things! Your design choice and justification is perhaps the best I have read in the whole class.

The only part that is lacking is the blueprint. You lost some points in that you did not include any numbers. Estimates would be absolutely fine in this case.